

STOCHASTIC METHODS IN WATER RESOURCES

Unit 4: Time series analysis

Lecture 11: Introduction and components of time series

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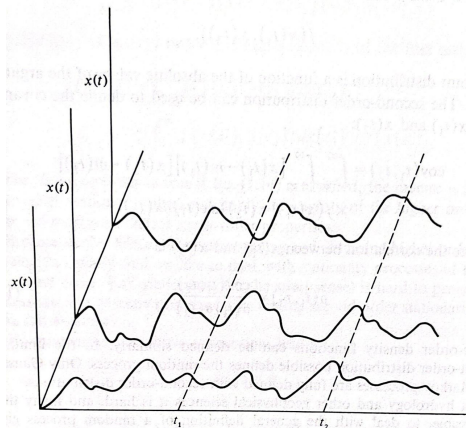
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Introduction

- ▶ The **analysis of time series** describe the **history of the movement in time** of some hydrological variables such as the flow rate in a particular section of a river.
- ▶ A **time series** is set of observations that measure the **variation in time** of certain phenomenon e.g. flow rate, water level in a lake.
- ▶ A **time series** pertains to the category **continuous state** and **discrete time** in which observations are made of continuously changing phenomena at **intervals of time** (usually equal) or an **average** or **total** value is taken for each time interval.
- ▶ **Air temperature** recorded each day at noon; mean monthly **barometric pressure** at a station.
- ▶ The information contained in a **discrete time series** is affected by the **sampling interval**.
- ▶ The interval depends on the purposes of the study. For instance when monitoring **water quality** in a stream hourly or daily samples are recommended, whereas for studying **water level** in a large reservoir, monthly data may suffice.
- ▶ Optimum sample intervals can be obtained based on **periodicity** and other phenomena characteristics.
- ▶ A essential characteristic of **time series** is **time-dependency** between observations. So, the order of the observations is important and the smaller the time interval the stronger the relationship between observations.

Introduction

- ▶ A **stochastic process**, $X(t)$, is a type of **random functions** defined in the **time** domain. The **stochastic process** is thus the collection of all possible **realizations** of $X(t)$. Each **realization** of $X(t)$ is **uncertain** and is known as **time series**, $x(t)$. In other way, $X(t)$ measures the **state** of the process at an particular time t .

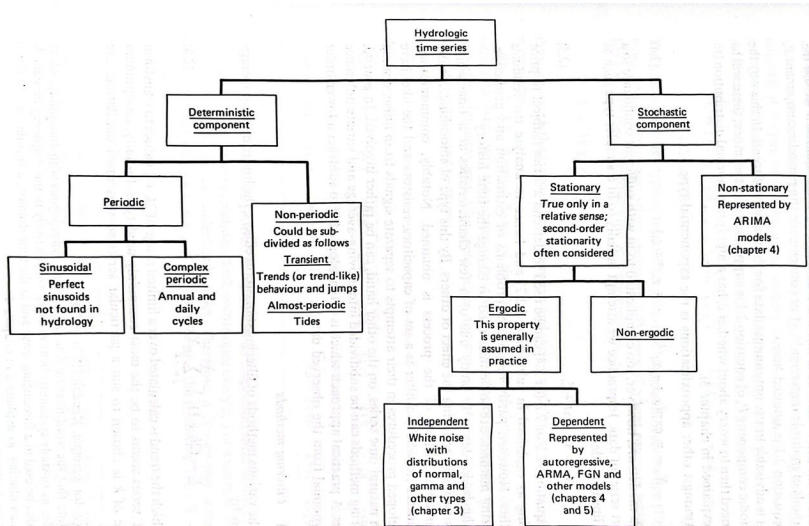


- ▶ The main objective in studying time series is:
 1. to **understand** the **mechanism** that generates the data. (most important)
 2. to **simulate** likely sequences of data over period o time.
 3. to **produce likely future sequences** or to **forecast** events over a short period of time.Note that **simulation** and **forecasting** is possible by making inferences regarding the underlying laws of the **stochastic process** from a sequence of observed data.

Components of time series

Generalities

The figure below shows the components and subdivisions of **hydrological time series**.



Components of time series

Generalities

Hydrological time series $x(t)$ can show different degrees of trend, jumps and displacements, and also stationarity, random fluctuations, autocorrelation and non Normality. Such components can be grouped in: **deterministic** and **stochastic**. While the **deterministic** components $d(t)$ are stationarity, trends, jumps and periodicities, the **stochastic** components $\xi(t)$ is constituted by random fluctuations that are not explained from the physical laws and thus require pdfs and autocorrelation function for their description. A $x(t)$ can be represented as

$$x(t) = d(t) + \xi(t)$$

where $d(t) = s(t) + b(t) + p(t)$ and $s(t)$, $b(t)$ and $p(t)$ represent trends, jumps and periodicities, respectively. Alternatively, $x(t)$ can be represented as

$$x(t) = d(t)\xi(t)$$

where $d(t) = s(t)b(t)p(t)$. This equation can be transformed into a lineal equation taken logarithms in both sides. The components of the time series are described below.

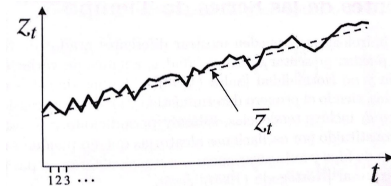
Components of time series

Trends

- ▶ **Trends** are continuous and gradual **increments** or **declines** of the hydrological variable during certain time period.
- ▶ **Trends** can occur during the whole time period or during part of the period.
- ▶ **Trends** can be attributed to **natural** or **anthropic** factors. For instance, **land-use changes**, **deforestation** and **climate change** in a basin trigger long-term trends in streamflow time series.
- ▶ In presence of **trends**, the time series shows a **positive** or **negative** correlation with time.

The following table show mathematical functions to represent **trends**.

Linear	$\bar{x}(t) = a + bt$
Polynomial	$\bar{x}(t) = a + bt + ct^2$
Power	$\bar{x}(t) = at^b$
Reciprocal	$\bar{x}(t) = \frac{1}{a+bt}$
Exponential	$\bar{x}(t) = ae^{-bt}$
Logistic	$\bar{x}(t) = \frac{a}{1+e^{-bt}}$



where $\bar{x}(t)$ is the mean of the hydrological time series, a , b and c are empirical constants, and t is the time. The figure shows an annual time series with a **positive trend** where the $\bar{x}(t)$ **increase linearly** with time.

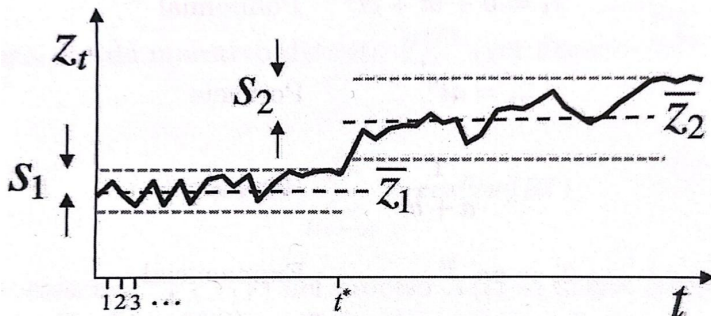
Components of time series

Jumps

Jumps are sudden changes in the **mean** and/or in the **variance** often caused by **natural disasters** such as earthquakes, landslides, volcano eruptions and fires. Such changes can be caused by **anthropic events** such as the construction of hydraulic structures (e.g. dams and diversions). **Jumps** can be mathematically represented by

$$\bar{x}(t) = \begin{cases} \bar{x}_1, & \text{if } t \leq t^* \\ \bar{x}_2, & \text{if } t > t^* \end{cases} \quad s(t) = \begin{cases} s_1, & \text{if } t \leq t^* \\ s_2, & \text{if } t > t^* \end{cases}$$

where $\bar{x}(t)$ and s_t are the **mean** and the **standard deviation** before and after of t^* (see the figure).



Note that the figure shows changes in the **mean** and in the **variance** due to a jump occurred at time t^* caused by a natural and/or anthropic event.

Components of time series

Trend and jump removal in annual series

Apart from assuming that the data set is **stationary** and **ergodic**, a time series must be free of **trends**, **jumps**, **periodicity**, etc prior being used. So that, the time series must be **transformed** in case such features are present in order to be used in modelling and forecasting. In consequence, **trends**, and **jumps** in **mean** and **variance** in time series with no seasonality can be removed as

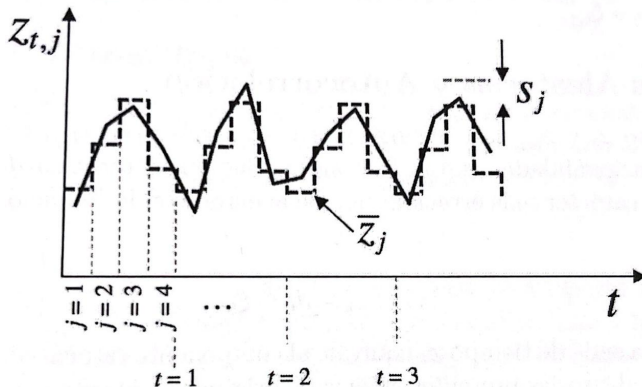
$$x'(t) = \frac{x(t) - \bar{x}(t)}{s(t)}$$

where $x(t)$ is the original series, $\bar{x}(t)$ is the **trend** of the **mean** represented by a function, or the **jump** in the **mean**, $s(t)$ is the **trend** in the **variance** represented by function (similar to the functions for the mean), or the **jump** in the **variance**, and $x'(t)$ is the resulting time series after removing **trends** and **jumps** in the **mean** and in the **variance**.

Components of time series

Periodicity

Sub-annual hydrological time series such as monthly time series can exhibit **periodic** or **cyclic** patterns as a consequence to the Earth translation around the Sun. This also drive the annual cycle of many hydrological processes. **Periodic** patterns can arise in sub-daily (e.g. hourly) time series due to the daily cycle due to the Earth rotation. Such **periodic** changes are presented in the **mean**, **variance**, **covariance** and **bias**. The figure below shows a three-month time series where one year is divided in 4 periods (e.g. seasons) of three months. Note that each period has its **mean** $\bar{z}_j$ and **standard deviation** s_j . The time series exhibit a **periodic** behaviour which is not presented in annual time series.



Components of time series

Periodicity

The **periodicity** in a time series can be represented by a set of sinusoidal functions. So, if the time series $x_j(t)$ does not have **trends** and **jumps**, just the periodicity, this can be represented as

$$x_j(t) = \bar{x} + \sum_{i=1}^L \lambda_i \sin \left\{ \frac{2\pi [(t-1)n + j]}{T} i + \phi_i \right\} + \xi_j(t)$$

where $\bar{x}$ is the **mean**, n is the number of periods is one year ($j = 1, \dots, n$), L is the number of **harmonics** (or **sinusoidal functions**) in one year, λ_i and ϕ_i are the **amplitude** and the **phase** of each harmonic i , and $\xi_j(t)$ is the random component. Regarding that $x_j(t)$ is discrete in time, if N is the length of the time series, the maximum number of harmonics is $L = N/2$ if N is even and $L = N/2 - 0.5$ if N is odd. In practice, the periodicity is usually represented between one and two harmonics in monthly time series and between four and six harmonics in daily time series.

Components of time series

Periodicity

Trends, jumps and periodicity removal in means and variances

Similarly to annual series with **trends** and **jumps**, these can be removed in series with periodicity. Since there are different **means** and **variances** for each period j of the year, such features can be removed as follow

$$x'_j(t) = \frac{x_j(t) - \bar{x}_j(t)}{s_j(t)}$$

where $x_j(t)$ is the original series, $\bar{x}_j(t)$ is the **mean** during the period j in t , $s_j(t)$ is the **standard deviation** in the period j in the time t , and $x'_j(t)$ is the resulting standardized time series. If the time series does not have neither **trends** nor **jumps** in the **means** and **variances**, $\bar{x}_j(t) = \bar{x}_j$ and $s_j(t) = s_j$, which means that each period j has constant **mean** and **variance** but different for each j . For a **periodic** time series with no **trends** and/or **jumps** in the **means** and **variances** in the j periods, a another way to remove the **periodicity** from the original series time series is substract the deterministic component as

$$x'_j(t) = x_j(t) - \bar{x} - \sum_{i=1}^L \lambda_i \sin \left\{ \frac{2\pi [(t-1)n + j]}{T} i + \phi_i \right\}$$

Note that in this equations $x'_j(t) \equiv \xi_j(t)$.

Components of time series

Random fluctuations and autocorrelation

If the series $x(t)$ and $x_j(t)$ are transformed into $x'(t)$ and $x'_j(t)$, respectively, the last ones are series without **trends**, **jumps** and **periodicity** (these three components constitute the deterministic component $d(t)$ of the time series) and are also **stationary**. Accordingly, the time series $x'(t)$ is

$$x'(t) = x(t) - d(t) = \xi(t)$$

which indicate that $x'(t)$ is equivalent to the random variations presented in the original time series $x(t)$. Similarly the time series $x'_j(t)$ is

$$x'_j(t) = x_j(t) - d_j(t) = \xi_j(t)$$

Note that $x'(t)$ and $x'_j(t)$ can have a structure of **temporal dependency** known as **autocorrelation** that establishes that the magnitude of the hydrological variable depends on its previous values, this means that the process has a form of '**memory**', frequently called **persistence**. The **persistence** indicates that low (high) values tend to be followed by low (high) values. In any **stochastic process** this temporal dependency is demonstrated through the **autocovariance function** or the **autocorrelation function**. For a **stationary** process, these two function depends only from the temporal lag τ .

Components of time series

Random fluctuations and autocorrelation

In the case of $x'(t)$, its linear temporal dependency structure is encapsulated in the **correlogram** (**autocorrelation function**), which is a plot of the **correlation** ρ_k vs the lag k . For the time series $x'_j(t)$, k represent the number of periods j in the years. For instance, for monthly time series it is possible to obtain 12 **correlograms**, one $\rho_{k,j}$ vs k relationship by month j . The coefficients $\rho_{k,j}$ correspond to the correlation of all values in j with all the values in the lagged month k . Note that when a **correlogram** decrease rapidly to zero after few k , this show a short **persistence** or **memory**. The opposite occurs when the **correlogram** decrease slowly.

Autocorrelation removal

The removal of the linear temporal dependency structure requires an mathematical model that represent the **correlogram**. The most simple model is a **autoregressive stochastic model** of lag 1. According to this model, the **annual transformed time series** can be expressed as

$$x'(t) = \rho_1 x'(t-1) + \varepsilon(t)$$

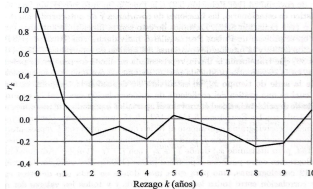
where ρ_1 is the **correlation coefficient** for lag $k = 1$ and $\varepsilon(t)$ is the **residual time series** with no **autocorrelation** (pure stochastic values). $\varepsilon(t)$ is represented by a **pdf** $f_\varepsilon(\epsilon)$ which is usually the **Normal distribution**.

Components of time series

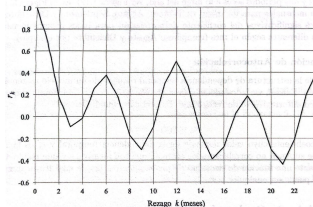
Random fluctuations and autocorrelation

The figures below show some **correlograms** for the streamflow data at the El Banco station in the Magdalena River.

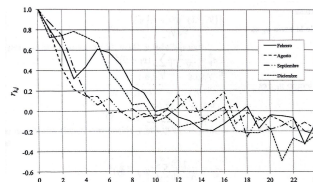
Correlogram of annual streamflows. ρ decay rapidly after $k > 0$, e.g. $\rho_1 = 0.14$.



Correlogram of monthly streamflows. There is a large dependency and show periodicity, e.g. $\rho_1 = 0.683$ explaining by the bimodal regime of the basin.

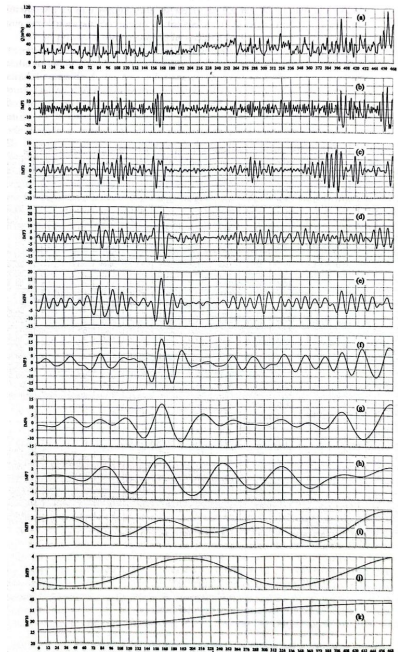


Monthly **correlograms** of streamflows. Strong dependency in the firsts k and the behaviour of the **correlogram** is different for each month.



Empirical modal decomposition

- ▶ Procedure developed by Huang et al 1998 that allow to decompose a time series in a finite number of **Intrinsic Modal Functions** with zero Mean.
- ▶ These functions are associated with different oscillatory modes embedded in the original series.
- ▶ The procedure is applied to **no linear** and **no stationary** time series.
- ▶ During the decomposition it is supposed that the series in any time t is composed of various oscillatory modes of different frequencies overlapping each other.
- ▶ The figure (panel a) shows the monthly streamflow in the Las Huertas station at the Bogota River. In the panel (d) it is possible to know the bimodal annual regime and in the panel (h) it is possible to detect the macro climatic ENSO signal.



Test for randomness and trend

In certain cases the presence of **trend** or **seasonality** is quite obvious, but often there is doubt whether the suspected effects are significant or not.

Turning point test

- ▶ This is a test for **randomness**.
- ▶ So in an observed sequence $x(t)$, $t = 1, 2, \dots, N$, a **turning point** p occurs at time $t = i$ if $x(i)$ is either greater than $x(i+1)$ and $x(i-1)$ or less than the two adjacent values.
- ▶ If we consider three unequal observations, the six possible orders of magnitude of these are: 1. $x(i-1) > x(i) > x(i+1)$, 2. $x(i-1) > x(i+1) > x(i)$, 3. $x(i) > x(i-1) > x(i+1)$, 4. $x(i) > x(i+1) > x(i-1)$, 5. $x(i+1) > x(i-1) > x(i)$ and 6. $x(i+1) > x(i) > x(i-1)$.
- ▶ In a random series, the six cases are equally probable and **turning points** occur in all except cases 1 and 6, which means that the chance of getting a **turning point** is $2/3$.
- ▶ Note also that, a **turning point** cannot occur when either $t = 1$ or $t = N$. So, the expected number of points in a random series is $E(p) = \frac{2(N-2)}{3}$ and the $Var(p) = \frac{16N-29}{90}$.
- ▶ p can be treated as an **standard measure** $z = \frac{p - E[p]}{(Var[p])^{1/2}}$, treated as an **standard normal deviate**. Note that p in z is the number of ps from the observations.
- ▶ Because too few or too many turning points indicate **non-randomness** (**null hypothesis**), a **two-tailed test of significance** is used.

Test for randomness and trend

Kendall's rank correlation test

- ▶ This test is based on the proportionate number of subsequent observations which exceed a particular value.
- ▶ For a sequence $x(1), x(2), \dots, x(N)$, the standard procedure is to determine the number of times, say, p , in all pairs of observations $(\{x(i), x(j)\}, j > i)$ that $x(j) > x(i)$.
- ▶ The ordered (i, j) subsets are $(i = 1, j = 2, 3, \dots, N), (i = 2, j = 3, 4, \dots, N) \dots (i = N - 1, j = N)$.
- ▶ Note that the **maximum possible number** of such pairs occurs for a continuously increasing sequence. This is **rising trend** where succeeding values are throughout greater than preceding ones and p is given by $(N - 1) + (N - 2) + \dots + 1$ which is the sum of an arithmetic progression given by $(N - 1)N/2$.
- ▶ If the observations are totally reversed, $p = 0$ and it follows that for a **trend-free series** $E[p] = N(N - 1)/4$.
- ▶ Note therefore that, if p is close to $(N - 1)N/2$ or 0, it indicates the presence of a **raising** or **falling** trend, respectively.
- ▶ The test is based on the $\tau = \frac{4p}{N(N-1)} - 1$ statistics for which $E[\tau] = 0$ and $Var[\tau] = \frac{2(2N+5)}{9N(N-1)}$, and that $\frac{\tau}{(Var[\tau])^{1/2}}$ converges rapidly to a **standard normal distribution** as N increases. Note that this relationship is thus the **standard test statistics**.
- ▶ The **null hypothesis** is of **no trend** in the series, and a **two-tailed test of significance** is used.

Test for randomness and trend

Regression test for linear trend

- ▶ This test is useful when it is thought that the **trend** is approximately **linear**. Standard methods of **linear regression** are used for this purpose.
- ▶ According to the expression for the **linear model** $X(t) = x_0 + \alpha t + \varepsilon(t)$, the hypothesis to be tested is $\alpha \neq 0$
- ▶ The first step is to estimate α and its variance denoted as $\hat{\alpha}$ and $s_{\hat{\alpha}}^2$, respectively.
- ▶ The statistics $t = \frac{\hat{\alpha}}{s_{\hat{\alpha}}}$ is then tested by using **Student's t test**. It is assumed here (as in other types of **regression analysis**) that the **residuals** $\varepsilon(t)$ are **stationary**, sequentially **independent** and **normally distributed**.
- ▶ The **null hypothesis** is of $\alpha = 0$ (**no trend**). So the statistics t is calculated and compared against the values from the **Student's t test**.

Other considerations:

- ▶ Note that **linearity** in **trend** is mainly a theoretical concept used to describe a general pattern, where **trends** in natural time series are **non-linear**.
- ▶ Also, **hydrologic time series** are not subject to **long-term movements**, generally speaking, over observed time spans. So, most of the exceptional cases appear to be caused by **local changes** in **physical conditions** (e.g catchment deforestation).

Moments and expectation

A single time series is considered a **stochastic process** that can be characterized by its (central) **statistical moments**. In particular, the first and second order moments are relevant: the **mean** value, the **variance** and the **autocorrelation** function. For an statistical **stationary process** the **mean** and the **variance** are:

$$\mu_X = E[X(t)] \quad \sigma_X^2 = \text{Var}[X(t)] = E[(X(t) - \mu_X)(X(t) - \mu_X)]$$

The **autocovariance** is a measure of the relationship of the process at two points in time. For two points in time k time steps apart, the **autocovariance** is defined by

$$\text{COV}[X(t), X(t+k)] = E[(X(t) - \mu_X)(X(t+k) - \mu_X)] \quad k = -\infty, \dots, -1, 0, 1, \dots, \infty$$

where k is often called **time lag**. Note that when $k = 0$ $\text{COV}[X(t), X(t+k)] = \sigma_X^2$ and $\text{COV}[X(t), X(t+k)] = \text{COV}[X(t+k), X(t)] = \text{COV}[X(t), X(t-k)]$

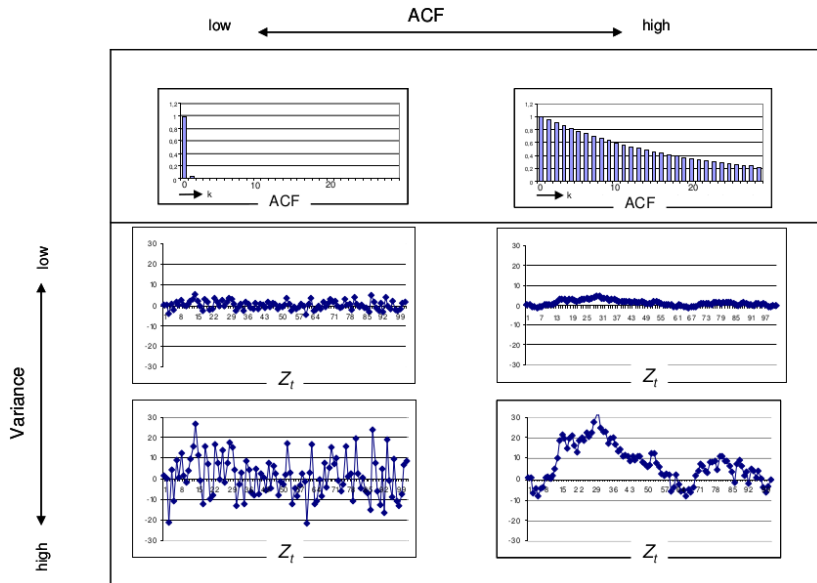
In **time series analysis** we often use the **autocorrelation function (acf)** defined by

$$\begin{aligned} \rho_{XX,k} &= \frac{\text{COV}[X(t), X(t+k)]}{\text{COV}[X(t), X(t)]} \\ &= \frac{E[(X(t) - \mu_X)(X(t+k) - \mu_X)]}{\sigma_X^2} \quad k = -\infty, \dots, -1, 0, 1, \dots, \infty \end{aligned}$$

Note that **acf** is always between 1 and -1, so a value of 1 or -1 **means a perfect correlation**, while a value 0 indicates the **absence of correlation**. From the definition it follows that the **acf** is maximum for $k = 0$ ($\rho_{XX,0} = 1$). Just like for the **autocovariance** it follows from the definition that $\rho_{XX,k} = \rho_{XX,-k}$
 $k = -\infty, \dots, -1, 0, 1, \dots, \infty$.

Moments and expectation

Different characteristics of zero mean time series due to high and low values for the **variance** and the **acf**. Note that the **dynamic behavior** of a time series is characterized by these two features.



Moments and expectation

Analogous to the autocovariance and the acf, that expresses the relation of a time series with itself, the relation between two different time series is expressed by the cross covariance and the crosscorrelation function (ccf). The cross covariance and ccf for the time series $X(t)$ and $Z(t)$ is defined as:

$$COV[X(t), Z(t+k)] = E [(X(t) - \mu_X)(Z(t) - \mu_Z)]$$

$$\rho_{XZ,k} = \frac{E [(X(t) - \mu_X)(Z(t) - \mu_Z)]}{\sigma_X \sigma_Z}$$

Analogous to the acf, the value of the ccf is always between 1 and -1. However the ccf is not symmetrical around $k = 0$: $\rho_{XZ,k} \neq \rho_{XZ,-k}$. However, from the definition of ccf $\rho_{XZ,k} \neq \rho_{ZX,-k}$.

Dealing with real data

Here we list some typical question that the analyst must ask itself for **cleaning** the time series, and to deal with **trend** and **seasonality**.

- ▶ Do you **understand** the context? Have the 'right' variables been **measured**?
- ▶ Have all the time series been **plotted**?
- ▶ Are there any **missing values**? If so, what should be done about them?
- ▶ Are there any **outliers**? If so, what should be done about them?
- ▶ Are there any obvious **discontinuities** in the data? If so, what does this mean?
- ▶ Does it make sense to **transform** any of the variables?
- ▶ Is **trend** present? If so, what should be done about it?
- ▶ Is **seasonality** present? If so, what should be done about it?